

Data in Brief: Four Benchmark Datasets for a Repeated-Evaluation Comparison of Traditional, Machine Learning, and Deep Learning Survival Models Across Static and Longitudinal Data

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Abstract

This article introduces four publicly available benchmark datasets used to evaluate traditional, machine learning, and deep learning survival models under a repeated-evaluation framework: Breast Cancer (SEER, $n = 4,024$), Heart Failure (Kaggle, $n = 299$), Recidivism (ICPSR study 8987, $n = 4,618$), and the longitudinal Mayo Clinic Primary Biliary Cirrhosis dataset (PBC2, 312 patients, 1,945 repeated observations). The first three datasets consist of static baseline covariates recorded at a single point in time, with event rates ranging from 15.3% to 37.3%, while PBC2 provides genuine time-varying covariate structure through repeated clinical measurements per patient (mean 6.2 visits, range 1–16). Together, the four datasets differ deliberately in structure, size, censoring rate, and clinical domain, enabling a rigorous test of whether model rankings among Cox Proportional Hazards (CoxPH), Random Survival Forest (RSF), and Long Short-Term Memory (LSTM) networks are stable across data conditions or depend on the presence of longitudinal structure. Models were evaluated using the concordance index (C-index), integrated Brier score (IBS), and time-dependent area under the curve (AUC) under a repeated train-test split protocol with post-hoc statistical testing. All four datasets are publicly available and support further research on survival model benchmarking, evaluation reliability, and the conditions under which recurrent architectures offer genuine predictive advantages.

Specifications Table

1 Value of the Data

- The four datasets jointly provide a deliberately heterogeneous benchmark spanning clinical oncology, cardiology, criminal justice, and hepatology domains, enabling assessment of model generalisability beyond a single application area.
- Sample sizes range from 299 to 4,618 and event rates from 15.3% to 37.3% across the three static datasets, supporting research on how dataset size and class imbalance affect the relative performance of traditional, machine learning, and deep learning survival models.
- The inclusion of PBC2 as a genuinely longitudinal dataset, alongside three static datasets, allows direct isolation of the effect of time-varying covariate structure on model ranking, in particular for recurrent architectures such as LSTM.

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Table 1: Specifications table.

Subject area	Biostatistics, Machine Learning
More specific subject area	Survival Analysis, Comparative Model Evaluation, Static and Longitudinal Data Modelling
Type of data	CSV files (three static tabular datasets; one longitudinal panel dataset)
How data were acquired	Publicly downloaded from SEER (Breast Cancer), Kaggle (Heart Failure), ICPSR (Recidivism), and the R <code>survival</code> package (PBC2)
Data format	Raw and preprocessed
Experimental factors	Class/event imbalance (event rates 15.3%–37.3% across static datasets); missing values in select covariates; longitudinal repeated-measures structure (PBC2 only)
Experimental features	Preprocessing: missing-value handling, encoding of categorical variables, censoring/event coding; Modelling: CoxPH, RSF, LSTM; Evaluation: C-index, Integrated Brier Score, time-dependent AUC, repeated train-test evaluation (100 iterations) with Shapiro–Wilk, Levene’s, ANOVA/Welch/Kruskal–Wallis, and Bonferroni-corrected post-hoc testing
Data source location	United States (all four datasets)
Data accessibility	See Data Availability section
Related research article	Tshisikule, O., Bere, A., & Mulaudzi, T.B. (2026). A Repeated-Evaluation Comparison of Traditional, Machine Learning, and Deep Learning Survival Models Across Static and Longitudinal Data. <i>Stats</i> (MDPI), submitted.

- The datasets enable a repeated-evaluation protocol (100 train-test splits per dataset) rather than single-split assessment, supporting research on the statistical reliability of survival model comparisons.
- All four datasets are open access, allowing full reproducibility of the companion study and reuse for further benchmarking research.

2 Data

2.1 Overview

The four datasets differ deliberately in structure, size, and clinical domain. Breast Cancer, Heart Failure, and Recidivism each consist of static baseline covariates recorded at a single point in time. PBC2, by contrast, is longitudinal: 312 patients contributed up to 16 repeated clinical measurements over time, providing genuine time-varying covariate structure.

Table 2: Dataset characteristics.

Characteristic	Breast Cancer	Heart Failure	Recidivism	PBC2
Sample size	4,024	299	4,618	312 patients
Observations	4,024	299	4,618	1,945 (longitudinal)
Number of events	616	96	1,721	140 deaths
Event rate (%)	15.3	32.1	37.3	44.9
Censoring rate (%)	84.7	67.9	62.7	55.1
Number of features	26	13	22	11
Data structure	Static	Static	Static	Longitudinal
Domain	Oncology	Cardiology	Criminal justice	Hepatology

2.2 Breast Cancer Dataset

Sourced from the SEER (Surveillance, Epidemiology, and End Results) Program of the National Cancer Institute (2017 November update), this dataset contains 4,024 breast cancer patient records with 26 features (23 categorical, 3 continuous), including patient demographics (age, race, marital status), tumour characteristics (TNM staging, differentiation grade, tumour size), hormone receptor status (estrogen and progesterone), lymph node examination results, and survival outcomes (survival months and vital status). This is the largest and most heavily censored dataset in the study (event rate 15.3%).

2.3 Heart Failure Dataset

Sourced from Kaggle, this dataset contains clinical records for 299 heart failure patients with 13 features (7 continuous, 6 categorical), including demographics (age, sex), clinical measurements (ejection fraction, serum creatinine, serum sodium, platelets, creatinine phosphokinase), comorbidities (anaemia, diabetes, high blood pressure), smoking status, follow-up time, and death events. This is the smallest dataset in the study and provides a test of model behaviour under limited sample size.

2.4 Recidivism Dataset

Sourced from ICPSR study 8987, this dataset contains 4,618 individuals with 22 features (5 continuous, 17 categorical), encompassing demographic variables, prior convictions, and employment indicators, with recidivism (return to custody) as the event of interest. This dataset has the highest event rate (37.3%) among the static datasets and extends the study beyond clinical domains into criminal justice.

2.5 PBC2 Dataset (Longitudinal)

The PBC2 dataset originates from a double-blind, placebo-controlled Mayo Clinic trial of primary biliary cirrhosis conducted between 1974 and 1984 [4], and is distributed as the `pbcseq` data object in the R `survival` package [5]. It contains 1,945 longitudinal observations from 312 patients, with a mean of 6.2 follow-up visits per patient (range 1–16). The outcome is recorded in the `status` variable: censored (143 patients, 45.8%), liver transplant (29 patients, 9.3%), or death (140 patients, 44.9%). Consistent with the death-only event definition adopted in the companion study, transplant recipients are treated as censored observations, yielding an overall censoring rate of 55.1% (45.8% true censoring plus 9.3% transplant) against an event (death) rate of 44.9%.

Sixteen predictor variables are recorded in total: treatment arm, age, and sex (static baseline covariates), visit day (the temporal index), histologic stage (a baseline covariate not included in the time-varying feature set), and 11 repeated clinical and biochemical measurements used as the LSTM’s time-varying sequence input: bilirubin, cholesterol, albumin, alkaline phosphatase, AST, platelets, prothrombin time, oedema, ascites, hepatomegaly, and spider angiomas. Serum cholesterol has the highest missingness among these (42.2%); all other variables, including the outcome, are fully observed. This is the only longitudinal dataset in the study and provides the time-varying covariate structure needed to test whether the LSTM’s architectural advantages are conditional on data structure.

2.6 Performance Metrics

Table 3: Performance metrics used for model evaluation.

Metric	Description
C-index	Concordance index; probability that predicted risk ordering matches observed event order
IBS	Integrated Brier Score; time-averaged squared error between predicted and observed survival
Time-dependent AUC	Area under the time-dependent ROC curve, evaluating discrimination at specific follow-up times

2.7 Software and Packages

Table 4: Software and packages used.

Purpose	Software/Package
Data manipulation	Pandas, NumPy
Visualisation	Matplotlib, Seaborn
Traditional survival modelling	lifelines (CoxPH)
Machine learning survival modelling	scikit-survival (Random Survival Forest)
Deep learning	TensorFlow/Keras (LSTM)
Statistical testing	SciPy (Shapiro–Wilk, Levene’s, ANOVA/Welch/Kruskal–Wallis, Bonferroni)
Programming language	Python 3.x

3 Results Summary

Single-split results, however, are known to be unreliable, particularly for high-variance models such as LSTM. Table 6 therefore reports the companion study’s primary evaluation: mean performance (standard deviation) across 100 independent repeated train-test splits per dataset.

3.1 Key Findings

1. On the three static datasets (Breast Cancer, Heart Failure, Recidivism), CoxPH and RSF consistently outperformed LSTM under repeated evaluation (mean C-index 0.66–0.73 vs. 0.31–0.42 for LSTM), with LSTM’s apparently competitive single-split performance (C-index 0.65–0.70) collapsing by 39.8–56.0% once evaluated across 100 splits.

Table 5: Single train-test split performance across all four datasets.

Dataset	Model	C-index	IBS	Time-Dep. AUC
Breast Cancer	CoxPH	0.762	0.048	0.713
	RSF	0.867	0.042	0.875
	LSTM	0.699	0.201	0.360
Heart Failure	CoxPH	0.737	0.123	0.669
	RSF	0.713	0.107	0.716
	LSTM	0.704	0.198	0.742
Recidivism	CoxPH	0.743	0.234	0.787
	RSF	0.708	0.201	0.749
	LSTM	0.646	0.227	0.385
PBC2 (Longit.)	CoxPH	0.728	0.168	0.764
	RSF	0.699	0.162	0.733
	LSTM	0.816	0.165	0.860

Table 6: Repeated-evaluation performance (100 iterations, mean (SD)) across all four datasets.

Dataset	Model	C-index	IBS
Breast Cancer	CoxPH	0.734 (0.016)	0.108 (0.003)
	RSF	0.721 (0.017)	0.110 (0.003)
	LSTM	0.421 (0.028)	0.229 (0.014)
Heart Failure	CoxPH	0.701 (0.043)	0.167 (0.016)
	RSF	0.727 (0.042)	0.162 (0.012)
	LSTM	0.310 (0.042)	0.205 (0.028)
Recidivism	CoxPH	0.659 (0.005)	0.229 (0.004)
	RSF	0.671 (0.012)	0.190 (0.004)
	LSTM	0.343 (0.014)	0.258 (0.010)
PBC2 (Longit.)	CoxPH	0.777 (0.042)	0.144 (0.022)
	RSF	0.779 (0.040)	0.135 (0.021)
	LSTM	0.806 (0.040)	0.169 (0.020)

2. On the longitudinal PBC2 dataset, this ranking reversed on discrimination: LSTM achieved the highest repeated-evaluation C-index of any model on any dataset in the study (0.806, SD 0.040), significantly outperforming CoxPH (0.777) and RSF (0.779), with only 1.2% degradation from its single-split estimate (0.816).
3. This reversal is not explained by sample size or censoring rate, both of which are broadly comparable to the other datasets. The distinguishing factor is structural: PBC2 is the only dataset with genuine repeated, time-varying measurements per subject, consistent with the interpretation that LSTM’s discrimination advantage is conditional on longitudinal data structure rather than a general property of the model class.
4. This advantage did not extend to calibration. Under repeated evaluation, LSTM recorded the poorest IBS on PBC2 (0.169) of the three models, behind RSF (0.135) and CoxPH (0.144),

indicating that LSTM’s superior ability to rank patients by relative risk on longitudinal data does not translate into better-calibrated absolute survival probability estimates.

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Data Availability

All datasets used in this study are publicly available:

- Breast Cancer dataset: Available on the [SEER Program of the National Cancer Institute \(2017 November update\)](#)
- Heart Failure dataset: Available on [Kaggle](#)
- Recidivism dataset: Available through [ICPSR](#)
- Mayo Clinic Primary Biliary Cholangitis Sequential Dataset (PBC2): Available through [the official R documentation page for the survival package](#)

All datasets and analysis code (including notebooks reproducing the single-split and 100-iteration repeated-evaluation results reported above) are additionally available at <https://github.com/justicetshisikule/Survival-Model-Comparison-with-Repeated-Evaluation-Across-Static-and-Longitudinal>

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